AFFIRMATIVE RESOLUTION OF ERDŐS PROBLEM 308 FOR EVERY $N \geq 1$

ILIA PAUZNER

For integers $N \geq 1$ and $k \geq 1$, write IsRepresentable(N, k) if there exists a subset $A \subseteq \{1, \dots, N\}$ such that $\sum_{n \in A} 1/n = k$, and put

$$\text{representable}(N) := \{k \in \mathbb{N} : k > 0 \text{ and } \text{IsRepresentable}(N, k)\}, \qquad m_N := \lfloor H_N \rfloor = \Big\lfloor \sum_{n=1}^{N} \frac{1}{n} \Big\rfloor.$$

In [2, p. 37] (see [1, 3, 4]), Erdős and Graham asked whether representable(N) has the shape $\{1, \dots, m\}$ for all sufficiently large N—settled asymptotically via Croot's bounds [1]—and whether the same holds unconditionally for *every* $N \geq 1$. We prove that the answer is yes.

Theorem. *For every integer $N \geq 1$, there exists $m \in \mathbb{N}$ (specifically $m \in \{m_N - 1, m_N\}$) such that* representable$(N) = \{1, \dots, m\}$.

Correspondence with the Lean formalization. The step-by-step proof below follows `FormalConjectures/ErdosProblems/308.lean`. All helper declarations live in a non-exposed private section inside `namespace Erdos308`, and the main theorem `erdos_308.variants.all_N` depends only on the standard Lean axioms {`propext`, `Classical.choice`, `Quot.sound`}.

This note	Lean (`Erdos308.*`)
IsRepresentable(N, k), representable(N), m_N	`IsRepresentable`, `representable`, `m`
Binary-tree evaluation on $[b, b + 2^d)$	`treeSum` (`treeSum_Ico`), `treeAll` (`treeAll_iff`)
Rational log and H_n bounds ($S_9 = 10^9$)	`logLb9`, `logUb9` (`log_bds`), `hCeil9` (`hCeil9_ok`)
Threshold sequence $W(k)$ and $H_{W(k)}$ bounds	`W`, `W_mono`, `W_bds`, `harm_W_bds`
Chebyshev ψ, θ bounds for $x \geq 300$	`psi_of_rows`, `cert_hi`, `psi_le_142`, `theta_bds`
Regime 1 ($1 \leq k \leq 9$) delta certificates	`treeLcm`, `bMask`, `chkCert`, `rep_W_le_9`
Modular gadget (M, D, S, A, B, η)	`Gadget`, `mkDivG`, `extendG`, `binGadget`, `autoG`
Prime-power denominator cancellation	`cancel_step`, `descend_sieve`, `peelPrime`
11520-smooth endgame and budget lemma	`rep_sieve`, `ValidPeel`, `rep_W_peel`
Prime-reciprocal partial summation	`inv_p_real`, `invDelta9_ok`, `invUb9_ok`, `inv_p_le`
Regimes 2–3 ($10 \leq k \leq 149$) row verifier	`chkRow`, `chkRow_ok`, `rW0`–`rWE`, `rep_W_10_149`
Regime 4 ($k \geq 150$) analytic proof	`qa_reg4_le`, `regime4_valid`
Main reduction for all $N \geq 1$	`all_N_main`, `erdos_308.variants.all_N`

1. Reduction to a Single Threshold Sequence $W(k)$

Define $W : \mathbb{N} \to \mathbb{N}$ by

$$W(0) := 0, \qquad W(1) := 1, \qquad W(2) := 10, \qquad W(k) := \lfloor e^{k+0.4188} \rfloor \quad (k \geq 3).$$

Lemma 1 (Threshold harmonic bounds; `harm_W_bds`, `all_N_main`). (1) *For every $k \geq 1$, $H_{W(k)} < k + 1$.*

(2) *For every $k \geq 5$, $k + 0.9958 < H_{W(k)} < k + 1$.*

(3) *Suppose* IsRepresentable$(W(k), k)$ *holds for every integer $k \geq 1$. Then for every $N \geq 1$,* representable$(N) = \{1, \dots, m\}$ *where $m = m_N$ if* IsRepresentable(N, m_N) *and $m = m_N - 1$ otherwise.*

Proof. For $1 \leq k \leq 4$, $W(k) \in \{1, 10, 30, 82\}$. Evaluating rational harmonic upper bounds (`hCeil9_ok`, derived from Padé bounds on $\log \frac{k+1}{k}$ and monotonicity of the Euler–Mascheroni sequence) gives $H_1 = 1 < 2$, $H_{10} < 2.93 < 3$, $H_{30} < 4.00 = 4$, and $H_{82} < 5.00 = 5$.

For $k \geq 5$, $W(k) \geq W(5) = 225 \geq 128$. Since $n \mapsto H_n - \log(n+1)$ is strictly increasing and $n \mapsto H_n - \log n$ is strictly decreasing on $\mathbb{N}_{>0}$ (`Real.strictMono_eulerMascheroniSeq`, `Real.strictAnti_eulerMascheroniSeq'`), bounding H_{128} to 10^{-9} precision yields for every $n \geq 128$:

$$\log(n+1) + 0.581119 < H_n < \log n + 0.581120.$$

From $e^{k+0.4188} < W(k)+1$ and $W(k) \leq e^{k+0.4188}$, we obtain $\log(W(k)+1) > k + 0.4188$ and $\log W(k) \leq k + 0.4188$, giving

$$k + 0.9958 < k + 0.999919 < H_{W(k)} < k + 0.999920 < k + 1.$$

For (3), every $k \in \text{representable}(N)$ satisfies $k = \sum_{n \in A} 1/n \leq H_N$, whence $k \leq \lfloor H_N \rfloor = m_N$. Conversely, if $1 \leq k \leq m_N - 1$, then $H_{W(k)} < k+1 \leq m_N \leq H_N$; strict monotonicity of H_n implies $W(k) < N$, so inclusion $\{1, \ldots, W(k)\} \subseteq \{1, \ldots, N\}$ lifts $\text{IsRepresentable}(W(k), k)$ to $\text{IsRepresentable}(N, k)$. Conditioning on $\text{IsRepresentable}(N, m_N)$ gives $\text{representable}(N) = \{1, \ldots, m_N\}$ or $\{1, \ldots, m_N - 1\}$. □

Everything therefore reduces to establishing $\text{IsRepresentable}(W(k), k)$ for every integer $k \geq 1$.

2. Effective Chebyshev and Prime-Reciprocal Bounds

Using $\log n = \sum_{d|n} \Lambda(d)$ in $\log(m!) = \sum_{n=1}^{m} \log n = \sum_{d=1}^{m} \Lambda(d) \lfloor m/d \rfloor$, any integer-weight stencil $\sum_i c_i \lfloor y/a_i \rfloor \geq 0$ with $\sum_i c_i/a_i = 0$ produces linear bounds on $\psi(x) = \sum_{n \leq x} \Lambda(n)$ via Stirling's inequalities

$$m \log m - m + \tfrac{1}{2} \log(2\pi m) \leq \log(m!) \leq m \log m - m + \tfrac{1}{2} \log(2\pi m) + \frac{1}{12m}$$

(`Stirling.le_log_factorial_stirling`, `Stirling.log_stirlingSeq'_antitone`). Applying a 39-term lower stencil (`tLoData`) and a 44-term upper stencil (`tHiNeg`) via `psi_of_rows`, checking $\operatorname{lcm}(1, \ldots, n)$ for $n \leq 299$ (`cert_hi`) to deduce $\psi(y) \leq 1.42y$ for all $y \geq 0$ (`psi_le_142`), and bounding $\psi(x) - \theta(x)$ via $\psi(x^{1/2}) + \psi(x^{1/3}) + \ldots$ (`psi_sub_theta_le_psi_add_psi_add_psi`) gives:

Lemma 2 (Effective θ and prime-reciprocal bounds)**.** (1) *For every real $x \geq 300$,*

$$0.9612\, x - 3.44 \log x - 4.26\sqrt{x} \leq \theta(x) \leq 1.0386\, x.$$

(2) *For integers $300 \leq a \leq b$, Abel summation on $\sum_{a < p \leq b} \frac{\log p}{p \log p}$ yields*

$$\sum_{a < p \leq b} \frac{1}{p} \leq \frac{0.0817\, a + 3.44 \log a + 4.26\sqrt{a}}{(a+1)\log(a+1)} + 1.0386 \log\Big(\frac{\log b}{\log a}\Big) + \frac{18.8}{a \log a}.$$

In particular, when $250{,}000 \leq a \leq b$, $\sum_{a < p \leq b} \frac{1}{p} \leq \frac{0.229 + 1.0386(\log b - \log a)}{\log a}$.

3. Regime 1 ($1 \leq k \leq 9$), Modular Gadgets, and Sieve Descent

Regime 1 ($1 \leq k \leq 9$). For each $1 \leq k \leq 9$, an explicit subset $A_k \subseteq \{1, \ldots, W(k)\}$ encoded as a delta list (`bMask`) is verified in exact integer arithmetic via $L_k = \operatorname{lcm}_{d \in A_k} d$ (`treeLcm`) to satisfy $\sum_{d \in A_k} (L_k/d) = kL_k$ (`chkCert`, `rep_W_le_9`).

Modular gadgets and prime-peeling descent ($k \geq 10$). Let $n = W(k) \geq 11520$ and choose $12 \leq M_{\text{top}} \leq M_{\text{bot}}$ with $M_{\text{top}} M_{\text{bot}} \leq n$. Define the target prime-power exponent cap

$$e_*(2) := 8, \qquad e_*(3) := 2, \qquad e_*(5) := 1, \qquad e_*(q) := 0 \quad (q \geq 7 \text{ prime}),$$

so $\prod_q q^{e_*(q)} = 2^8 \cdot 3^2 \cdot 5 = 11520$. Starting from $\{1, \ldots, n\}$, form the sieved set S_0 by discarding:

(1) multiples of primes $p > n/M_{\text{top}}$, with reciprocal sum bounded by

$$\mathcal{L}_{\text{tail}}(M_{\text{top}}, n) := \sum_{m=1}^{M_{\text{top}}-1} \frac{1}{m} \sum_{n/M_{\text{top}} < p \leq n/m} \frac{1}{p};$$

(2) multiples of p^2 for primes $M_{\mathrm{bot}} < p \le n/M_{\mathrm{top}}$, bounded by $\mathcal{L}_{\mathrm{sq}}(M_{\mathrm{bot}}, n) := \sum_{M_{\mathrm{bot}}<p\le\sqrt{n}} \frac{H_{\lfloor n/p^2\rfloor}}{p^2}$;
(3) multiples of q^a ($a \ge e_*(q)+1$) for primes $q \le M_{\mathrm{bot}}$, bounded by $\mathcal{L}_{\mathrm{QA}}(M_{\mathrm{bot}}, n)$ using $\sum_{a\ge 9} 2^{-a}$ at $q=2$ and, for odd $q \le M_{\mathrm{bot}}$ with $k_q := \lfloor \log_2 q\rfloor + 1$, the dyadic bound $(2^{k_q}-1)/(q^a \cdot 2^{\lfloor \log_2(n/q^a)\rfloor})$ when $q^a \cdot 2^{k_q-1} \le n$ (removing only 2-power-free cofactor multiples via binary subset sums, `sub_two_pow`) and $H_{\lfloor n/q^a\rfloor}/q^a$ otherwise (`descend_sieve`).

A *modular gadget* (`Gadget`) is a tuple $G = (M, D, S_G, A, B, \eta)$ where $D \ge 1$ is M-smooth, $S_G \subseteq \{1, \dots, M\}$ consists of divisors of D, $\eta = \sum_{m\in S_G} 1/m$, and every integer $t \in [A, B]$ is a subset sum $t = \sum_{m\in T} D/m$ ($T \subseteq S_G$). Two constructions supply all gadgets needed:

- **Greedy residue-extension gadgets (`mkAutoSpec`, `autoG`)**: starting from exact divisor DP on $D_0 = 2520$ (`mkDivG`), repeatedly multiplying $D \mapsto Dq$ and choosing via `pickTq` a subset T_q of divisors of Dq in $\{1, \dots, M\}$ not dividing D whose scaled reciprocals $(Dq)/m$ span all q residues modulo q, lifting $[A, B] \mapsto [qA + s_q, qB]$ (`extendG`).
- **Binary-lifting prime-product gadgets (`binGadget`)**: for any $R \ge 3$, taking $M = 2R^2$ and $D = 2^{1+\lfloor \log_2 R\rfloor} \prod_{3\le q\le R} q$, the powers of 2 up to $D_0 = 2^{1+\lfloor \log_2 R\rfloor}$ cover $[1, D_0 - 1]$, and adjoining $\{q \cdot 2^j : 0 \le j \le \lfloor \log_2 q\rfloor\}$ for each odd prime $q \le R$ covers all residues modulo q via $D_0^{-1} \bmod q$ (`ZMod.unitOfCoprime`), maintaining $B - A \ge D$ and $\eta \le 1$.

Lemma 3 (Prime peeling and 11520-smooth completion; `peelPrime`, `rep_sieve`, `rep_W_peel`)**.** *Suppose for every prime $p \in (M_{\mathrm{bot}}, n/M_{\mathrm{top}}]$ we assign a gadget G_p with $G_p.M \le M_{\mathrm{bot}}$, $p \cdot G_p.M \le n$, and $G_p.B - G_p.A + 1 \ge p$. Let $k_p := \min(G_p.M, \lfloor n/(p \cdot G_p.M)\rfloor) \ge 1$ and*

$$\mathcal{L}_{\mathrm{peel}}(M_{\mathrm{top}}, M_{\mathrm{bot}}, n, G) := \sum_{M_{\mathrm{bot}}<p\le n/M_{\mathrm{top}}} \frac{G_p.\eta + (p-1)/G_p.D}{p\, k_p}.$$

If $\mathcal{L}_{\mathrm{tail}}(M_{\mathrm{top}}, n) + \mathcal{L}_{\mathrm{sq}}(M_{\mathrm{bot}}, n) + \mathcal{L}_{\mathrm{peel}}(M_{\mathrm{top}}, M_{\mathrm{bot}}, n, G) + \mathcal{L}_{\mathrm{QA}}(M_{\mathrm{bot}}, n) \le 0.9921$, then IsRepresentable$(n, k)$ *holds (for $n = W(k)$, $k \ge 5$).*

Proof. For each prime $p \in (M_{\mathrm{bot}}, n/M_{\mathrm{top}}]$, every element pk_pm ($m \in G_p.S$) lies in S_0 and has no prime factor $> M_{\mathrm{bot}}$ other than p[1] (so the sets $\{pk_pm : m \in G_p.S\}$ are pairwise disjoint across distinct p). Since $G_p.D$ is M_{bot}-smooth and $p > M_{\mathrm{bot}}$ is prime, $\gcd(p, G_p.D) = 1$. Multiplying the current p-residue of the numerator by $(G_p.D)^{-1} \bmod p$ selects a unique $t_p \in [G_p.A, G_p.A + p - 1] \subseteq [G_p.A, G_p.B]$ such that deleting $\{pk_pm : m \in T_p\}$ ($\sum_{m\in T_p} G_p.D/m = t_p$) eliminates p from the reduced denominator (`cancel_step`, `peelPrime`) at reciprocal cost $t_p/(G_p.D \cdot pk_p) \le (G_p.\eta + (p-1)/G_p.D)/(pk_p)$.

After peeling all $p \in (M_{\mathrm{bot}}, n/M_{\mathrm{top}}]$, every prime power dividing the reduced denominator of $\sum_{d\in S_1} 1/d$ has exponent $\le e_*(q)$, so $\sum_{d\in S_1} 1/d = u/11520$ for some $u \in \mathbb{N}$ (`sub_fac_le`). By Lemma 1(2), $H_n > k + 0.9958$, so subtracting ≤ 0.9921 leaves $k < u/11520 \le H_n < k + 1$, i.e. $0 < u - 11520k < 11520$. All 54 divisors of 11520 remain in S_1, and subset-sum DP over those 54 divisors represents every integer in $[0, 42466]$ as $\sum_{d\in U} 11520/d$ (`rep_sieve`). Deleting U for $r = u - 11520k < 11520 \le 42466$ yields $\sum_{d\in S_1\setminus U} 1/d = k$. □

4. Closing $10 \le k \le 149$ and $k \ge 150$

Regimes 2–3 ($10 \le k \le 149$; `rW0`–`rWE`, `rep_W_10_149`). We partition $[10, 149]$ into 15 intervals ($[10,10], \dots, [18,18]$, $[19,21]$, $[22,25]$, $[26,34]$, $[35,55]$, $[56,99]$, $[100,149]$). On each interval $[k_{\mathrm{lo}}, k_{\mathrm{hi}}]$, certified enclosures $n_{\mathrm{lo}} \le W(k) \le n_{\mathrm{hi}}$ (`W_bds`) and a two-gadget step assignment $G_p = G_0$ for $p > \lfloor n/M_1^2\rfloor$ and $G_p = G_1$ for $p \le \lfloor n/M_1^2\rfloor$ ($s_i =$ `mkAutoSpec`(M_i, R_i)) are checked by the 10^{-9}-scaled integer verifier `chkRow`. Kernel evaluation (`by decide` in `rW0`–`rWE`) verifies `rowUb9` $\le 992{,}100{,}000$, whence `chkRow_ok` yields IsRepresentable$(W(k), k)$ for all $10 \le k \le 149$.

Regime 4 ($k \ge 150$; `qa_reg4_le`, `regime4_valid`). For $k \ge 150$, set $R := \lfloor 4k/3\rfloor + 220 \ge 420$ and let G be the binary-lifting gadget (`binGadget`) at R, with $M_{\mathrm{top}} = M_{\mathrm{bot}} = M := 2R^2 \in [352{,}800,\ 16k^2]$, $840k \le M$, $\eta \le 1$, and $G.D \ge 2\exp(\theta(R))$. By Lemma 2(1), $\theta(R) > k + 11$, so $G.D > 1000e^{k+1} > 1000n$ ($n = W(k) < e^{k+1}$), giving $(p-1)/G.D < 0.001$ and $G.B -$

$G.A \geq G.D > n$. Moreover $4M^6 \leq 4(16k^2)^6 < e^k < n$, so $B := \lfloor n/M^2 \rfloor \geq 250{,}000$ and $a_{\text{tail}} := \lfloor n/M \rfloor \geq B$. Setting $x := k/150 \geq 1$ gives $\log M \leq 13.17\sqrt{x}$, $(\log M)^2 \leq 173.5x$, $\log a_{\text{tail}} \geq 136.8x$, and $\log B \geq 123.65x$. Bounding the four loss components via Lemma 2(2):

(1) $\mathcal{L}_{\text{tail}}(M,n) \leq \dfrac{(0.229 + 1.0386/250000)(1+\log M) + 1.0386(\log M + \frac{1}{2}(\log M)^2)}{\log a_{\text{tail}}} \leq 0.790$ (using $\sum_{m=1}^{M-1} \frac{1}{m} \log \frac{M}{m} \leq \log M + \frac{1}{2}(\log M)^2$, `inv_log_le`);

(2) $\mathcal{L}_{\text{sq}}(M,n) \leq \dfrac{H_{\lfloor n/(M+1)^2 \rfloor}}{M} < \dfrac{k+1}{840k} \leq 0.003$;

(3) splitting $\mathcal{L}_{\text{peel}}(M,M,n,G)$ at $B = \lfloor n/M^2 \rfloor$: on $(M,B]$ we have $k_p = M$ and $\sum_{M<p\leq B} \frac{1.001}{Mp} \leq \frac{1.001}{M} \cdot 0.2k \leq 0.001$, while on $(B, a_{\text{tail}}]$ we have $k_p \geq 1$, $\log a_{\text{tail}} - \log B \leq 13.18x$, and $\sum_{B<p\leq a_{\text{tail}}} \frac{1.001}{p} \leq 1.001 \times 0.113 < 0.114$, giving $\mathcal{L}_{\text{peel}} \leq 0.115$;

(4) $\mathcal{L}_{\text{QA}}(M,n) \leq 0.024$ (`qa_reg4_le`): powers of 2 ($a \geq 9$) contribute $\leq 1/256 < 0.004$; for odd primes $3 \leq q \leq M$, exponents $1 \leq a \leq 5$ satisfy $q^a \cdot 2^{\lfloor \log_2 q \rfloor} \leq M^6 < n/4$ so each contributes $\leq 4M/n$ (at most $20M^2/n \leq 0.001$ in total), and exponents $a \geq 6$ contribute $\leq \sum_{q=3}^{M} \sum_{a\geq 6} q^{1-a} \leq \frac{1}{27} \sum_{q=3}^{M} \frac{1}{q(q-1)} \leq \frac{1}{54} < 0.019$.

Summing gives $\mathcal{L}_{\text{tail}} + \mathcal{L}_{\text{sq}} + \mathcal{L}_{\text{peel}} + \mathcal{L}_{\text{QA}} \leq 0.790 + 0.003 + 0.115 + 0.024 = 0.932 \leq 0.9921$. Lemma 3 yields IsRepresentable$(W(k), k)$ for all $k \geq 150$, and Lemma 1(3) completes the proof of the Theorem. □

References

[1] E. S. Croot, III, *On some questions of Erdős and Graham about Egyptian fractions*, Mathematika **46** (1999), 359–372.

[2] P. Erdős and R. L. Graham, *Old and new problems and results in combinatorial number theory*, Monographies de L'Enseignement Mathématique, vol. 28, Université de Genève, 1980.

[3] *Erdős Problem #308*, `https://www.erdosproblems.com/308`.

[4] The Formal Conjectures Authors, *FormalConjectures/ErdosProblems/308.lean*, `https://github.com/google-deepmind/formal-conjectures`.